

EXECUTIVE SUMMARY · ENTERPRISE READINESS

From prototype to production.

This document defines the exact artifacts, infrastructure, compliance, and team milestones required to move SCENTS from a working local prototype to a production-grade enterprise platform.

[View roadmap](#)[Artifacts built](#)

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CURRENT STATE

What is built and verified today.

Core Platform

54/54 tests

28 modules implemented and wired: ingestion, transform, ML, JanusGraph, SAIM loop, SERL entity resolution, explainability, fusion, causal, privacy, security, consolidation, data products, monitoring.

WISMO Operational Layer

24/24 tests

8 backend services running: gateway :3000, order :3001, geo :3002, notification :3003, are :3004, route :3005, pixel :3007, delphi :5025. Role-gated merchant and rider dashboards.

Investor Artifacts

3 live demos

Public landing page, investor pitch, Family Bank pitch, and auto-sync deployment to GitHub Pages. All dashboards serve locally and via ekn1/DEMO.

Verified Behaviors

CAPABILITY	STATUS
Order creation and persistence	● verified
CORS preflight on order service	● 204 verified
Rider login gate and role restriction	● verified
Pixel event ingestion	● 200 verified
OSINT connectors	● 24/24 tests
Endpoint health checks	● 8/8 services
GitHub Pages deployment	● live

GAPS

What is missing for enterprise readiness.

Infrastructure

missing

No cloud hosting, no Kubernetes, no Terraform, no CI/CD pipelines, no container registry, no load balancers, no DNS, no TLS automation.

Data Stack

partial

JanusGraph exists locally. No production HA graph cluster, no Druid/ClickHouse OLAP, no Spark batch layer, no Data Lake with retention policies, no GraphQL mutation API.

Security & Compliance

missing

No ISO 27001/SOC 2 implementation, no pen-testing, no secrets management, no WAF, no DDoS protection, no Kenya Data Protection Act 2019 compliance framework, no client data processing agreements.

Team & Operations

missing

No CTO, no engineers, no DevOps, no SRE, no security engineer, no legal/compliance counsel, no commercial team. No 24/7 incident response, no monitoring, no SLOs.

ENTERPRISE ARTIFACTS

Artifacts produced in this session.

The following files constitute the enterprise foundation. They are production-ready scaffolding that a deployed team can extend immediately.

Containerization

docker

Production Dockerfile for SCENTS API with non-root user, health checks, and uvicorn. Docker Compose file with JanusGraph, Kafka, Postgres, Redis, SCENTS API, and all WISMO services.

deploy/docker/Dockerfile.scents-api
deploy/docker/docker-compose.yml

Kubernetes Manifests

k8s

Namespace, ConfigMap, Secret template, Deployment with HPA, Service, and security context for SCENTS API. Full set of Deployments and Services for all WISMO services with 2-3 replicas each.

deploy/k8s/scents-api/
deploy/k8s/wismo/
deploy/k8s/dashboard/

Infrastructure as Code

terraform

AWS EKS cluster, VPC with public/private subnets, NAT gateways, EKS node group with autoscaling. Staging and prod environment

CI/CD Pipeline

github actions

Automated test → build → deploy workflow. Pushes to develop deploy to staging; pushes to

configs. Outputs for kubeconfig and cluster endpoint.
deploy/terraform/modules/scents/
deploy/terraform/environments/{staging,prod}/

main deploy to prod with rollout status check.
Requires kubeconfig secrets.
.github/workflows/scents-ci.yml

NEXT STEPS

Phased enterprise readiness plan.

PHASE 1 FOUNDATION

Weeks 1–4: Cloud foundation and container registry

- Provision AWS account and enable EKS in af-south-1
- Run Terraform to create VPC, subnets, NAT, EKS cluster, node group
- Create ECR repositories for scents-api, wismo-gateway, wismo-* services, dashboard
- Push Docker images to ECR and verify deployments in staging namespace
- Set up external DNS and TLS via cert-manager + Route53
- Enable CloudWatch logging for cluster and node groups

PHASE 2 DATA STACK

Weeks 5–8: Production data services

- Deploy RDS Postgres with Multi-AZ and automated backups
- Deploy ElastiCache Redis cluster with replication
- Deploy JanusGraph on EC2/EKS with persistent EBS volumes
- Deploy MSK (managed Kafka) with topic configs and retention policies
- Deploy Druid or ClickHouse for real-time OLAP
- Implement S3-based Data Lake with lifecycle policies
- Build GraphQL mutation API for Delphi and knowledge graph writes

PHASE 3 SECURITY

Weeks 9–12: Security and compliance baseline

- Implement Vault or AWS Secrets Manager for all credentials
- Enable network policies, WAF, and security groups
- Run initial pen-test and remediate critical findings
- Draft Data Protection Act 2019 compliance documentation
- Implement audit logging and tamper-evident storage
- Prepare client data processing agreements and DPAs

- Enable MFA for all production access

PHASE 4 OBSERVABILITY

Weeks 13–16: Monitoring, alerting, and SLOs

- Deploy Prometheus + Grafana or AWS Managed Prometheus
- Define SLOs: availability, latency, throughput, error rate
- Set up PagerDuty or Opsgenie for on-call rotation
- Implement distributed tracing (OpenTelemetry)
- Build runbooks for common failure modes
- Conduct first incident response drill

PHASE 5 TEAM & COMMERCIAL

Weeks 17–24: Hiring and client onboarding

- Hire CTO/VP Engineering and first 2 backend engineers
- Hire DevOps/Platform Engineer and Cybersecurity Engineer
- Hire Head of Commercial and 2 Client Intelligence Managers
- Onboard first 3 enterprise clients under NDA
- Conduct ISO 27001 gap assessment with auditor
- Launch Insights API and Intelligence Suite in staging

TEAM

Critical roles to hire before production launch.

CTO / VP Engineering

1

Own full technical architecture. Must have Kafka, Spark, JanusGraph/Neo4j, Kubernetes. Recruit first.

Senior ML Engineer

2

Own Scent Classifier, entity extraction, relationship scoring. Knowledge graphs and NLP required.

Data Engineer

2

Own ingestion services, Kafka topics, transformation services, Data Lake. Kafka, Spark, Druid/ClickHouse.

Backend Engineer

2

Own BFF Service and client-facing API layer. Node.js or Python, GraphQL, Postgres, Redis.

DevOps / Platform Engineer

1

Kubernetes, Terraform, AWS/GCP, CI/CD, data residency, infrastructure security hardening.

Cybersecurity Engineer

1

Threat modeling for data-intensive platforms, pen-testing, ISO 27001 or SOC 2 implementation.

Legal & Compliance Counsel

retainer

Kenya Data Protection Act 2019, financial sector regulations, client DPAs, cross-border data governance.

Head of Commercial

1

7+ years B2B tech sales in Kenya financial services or FMCG. Own enterprise pipeline and API partnerships.

Client Intelligence Manager

2

Enterprise account management. Translate platform outputs into strategic recommendations for C-suite.

INVESTMENT

Estimated 6-month burn to production-ready.

Infrastructure

monthly

EKS cluster, RDS, ElastiCache, MSK, S3, CloudWatch, Route53, ECR, NAT gateways, data transfer: ~\$8,000–\$15,000/month depending on load and data volume.

Personnel

6 months

CTO, 2 ML, 2 Data, 2 Backend, DevOps, Cybersecurity, Legal retainer, Head of Commercial, 2 Client Managers: ~\$500,000–\$800,000 total compensation at market rates.

Security & Compliance

one-time

Pen-test, ISO 27001 gap assessment, SOC 2 readiness, legal documentation: ~\$50,000–\$100,000.

Data Partnerships

variable

CRB, telco, agricultural, government API access: negotiable, often revenue-share or upfront licensing.

RISKS & MITIGATIONS

Key enterprise deployment risks.

RISK	LIKELIHOOD	IMPACT	MITIGATION
Data residency / cross-border transfer restrictions	● high	high	Kenya-hosted nodes, in-region processing, legal review before any cross-border flow
Talent scarcity for specialized roles	● high	high	Remote-first hiring, competitive global rates, Nairobi hub, contractor bridge
Client data leakage or breach	● medium	critical	Zero-trust architecture, encryption at rest and in transit, pen-testing, insurance
Regulatory change invalidating assumptions	● medium	high	Legal retainer, modular compliance layer, agile policy updates
Dependency on WISMO seed data	● medium	medium	Synthetic data fallback, parallel data partnerships, phased onboarding

SUCCESS METRICS

How we measure enterprise readiness.

Technical

- 99.9% API availability over 30 days
- P99 latency <200ms for inference endpoints
- All 78 tests passing in CI/CD
- Zero critical vulnerabilities in pen-test

Commercial

- 3 enterprise clients onboarded
- Insights API at 10+ developer signups
- WISMO integration premium revenue active
- Data partnership agreements signed

Compliance

- ISO 27001 certification in progress
- DPA templates reviewed by counsel
- Data retention policies enforced
- Audit trail completeness verified

DECISION

The ask.

The prototype is built, tested, and demonstrable. The enterprise artifact set is complete. What remains is execution: cloud provisioning, hiring, security hardening, and client onboarding.

We are seeking strategic investment and partnership to fund Phase 1–3 and assemble the core team.

[Request technical briefing](#)[Discuss investment](#)

ekn1/DEMO	ekn1/wismo
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